

Multiple Choice (10 marks)

1. Suppose today is Wednesday. What day of the week will it be $10^{(10^{(10)})}$ days from now?
 2. Sunday
 3. Monday
 4. Tuesday
 5. Wednesday

How many real roots does the polynomial $2x^5 + 8x - 7$ have?

- (a) None
- (b) One
- (c) Two
- (d) Three

Compute the product in the given ring. $(20)(-8)$ in $\mathbb{Z}_{26}$

- (a) 0
- (b) 1
- (c) 11
- (d) 22

Statement 1 | For $n > 1$, the set $\{1, 2, \dots, n-1\}$ is a group under multiplication modulo n .

Statement 2 | There is an integer x such that $63x \bmod 100 = 1$.

- (a) True, True
- (b) False, False
- (c) True, False
- (d) False, True

$(\mathbb{Z}, *)$ is a group with $a*b = a+b+1$ for all a, b in $\mathbb{Z}$. The inverse of a is

- (a) 0
- (b) -2
- (c) $a-2$
- (d) $(2+a)^{-1}$

True / False (10 marks)

Answer True or False

6. True or False: Not every maximal ideal is a prime ideal; maximal ideals are prime only in certain types of rings.
7. True or False: For $c = 0$ in $\mathbb{Z}_3$, the polynomial $x^3 + cx^2 + 1$ is irreducible, making $\mathbb{Z}_3[x]/(x^3 + cx^2 + 1)$ a field.
8. True or False: If H is a subgroup of G and a belongs to G , then the left coset $|aH|$ is equal to the right coset $|Ha|$.
9. True or False: The condition that any two distinct column vectors of M are linearly independent is not equivalent to the other conditions on M .
10. True or False: The dimension of the intersection of two 4-dimensional subspaces in a 7-dimensional vector space can be 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In $\triangle ABC$, shown, $\cos B = \frac{3}{5}$. What is $\cos C$?
[asy] draw((0,0)--(4,0)--(0,3)--cycle,black+linewidth(1));
draw(rightanglemark((4,0),(0,0),(0,3),10),black+linewidth(1)); label("A",(0,0),W); label("B",(0,3),W); label("C",(4,0),E); label("90°",(0,0)--(0,3),W); [/asy]
12. What is the value of b if $-x^2 + bx - 5 < 0$ only when $x \in (-\infty, 1) \cup (5, \infty)$?

End of Paper